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SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

MINI PROJECT REPORT

PERFORMANCE ANALYSIS OF A 40 GBPS CWDM-BASED WDM OPTICAL SYSTEM: A COMPARATIVE STUDY OF PIN AND APD PHOTODETECTORS

ETEG 413: Optical Fiber Communication

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ABSTRACT

This report presents the design, simulation, and performance evaluation of a four-channel Wavelength Division Multiplexing (WDM) optical communication system implemented using Coarse Wavelength Division Multiplexing (CWDM) components and fiber operating at 40 Gbps, implemented using OptiSystem. The system multiplexes four channels at 1310 nm, 1311.6 nm, 1313.2 nm, and 1314.8 nm using a WDM Mux 4x1, transmits the combined signal over a single-mode CWDM fiber with an Erbium-Doped Fiber Amplifier (EDFA) compensating span loss, and recovers each channel using a WDM Demux 1x4 followed by independent photodetection and 3R regeneration stages. The primary objective of this study is to compare the performance of PIN and Avalanche Photodiode (APD) receivers across a fiber length sweep from 0 km to 100 km, using Q-factor, Bit Error Rate (BER), and eye diagram quality as the performance metrics. Simulation results show that the PIN receiver generally achieves a higher Q-factor at shorter transmission distances, while the APD receiver becomes advantageous at the maximum investigated distance of 100 km as reduced received power makes its internal gain increasingly beneficial. At the system's validated baseline length, Q-factor values of 7.60 to 11.82 were achieved with corresponding BER values below 1.53×10^{-14}, confirming reliable, near-error-free transmission. These findings provide practical guidance on photodetector selection for CWDM-based metro and access network deployments depending on the required transmission reach.

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INTRODUCTION

1.1. Background

The continuous growth in data traffic driven by cloud computing, video streaming, and enterprise networking has placed increasing demand on the capacity of optical transmission systems. WDM addresses this demand by allowing multiple independent data channels, each carried on a distinct optical wavelength, to be transmitted simultaneously over a single optical fiber. Within the WDM family, CWDM offers a cost-effective alternative to Dense WDM (DWDM) by using wider channel spacing, which relaxes the requirements on laser temperature stabilization and filter precision, making it well suited to metropolitan and access network applications where channel count requirements are moderate and cost is a primary constraint.

This study designs and simulates a four-channel WDM system implemented using CWDM components, operating at an aggregate data rate of 40 Gbps using OptiSystem. Beyond verifying basic system functionality, the project places particular emphasis on comparing two common photodetector technologies, PIN photodiodes and Avalanche Photodiodes (APDs), across a range of transmission distances. PIN photodiodes are simple, low-noise devices with no internal gain, while APDs use avalanche multiplication to provide internal gain at the cost of additional multiplication noise. Understanding how these two technologies perform as a function of fiber length is essential for selecting the appropriate receiver technology for a given network reach.

1.2. Objectives

The objectives of this project are:

- To design and simulate a four-channel, 40 Gbps CWDM-based WDM optical communication system.
- To evaluate system performance using spectral analysis, Q-factor, and BER.
- To compare PIN and APD receiver performance over 0–100 km fiber lengths.

TECHNICAL PRINCIPLES

2.1. CWDM Fundamentals

Wavelength Division Multiplexing (WDM) combines multiple optical signals of different wavelengths onto a single fiber, multiplying its usable capacity without added physical infrastructure. WDM systems are classified as Coarse WDM (CWDM) or Dense WDM (DWDM) based on channel spacing. CWDM, standardized by the ITU-T G.694.2 grid, typically uses 20 nm spacing across the 1270–1610 nm range, enabling uncooled lasers and simpler filters at lower system cost. DWDM uses much narrower spacing (0.4–0.8 nm, or 50–100 GHz) to pack more channels at the cost of more complex, expensive components.

This study uses four channels centered at 1310, 1311.6, 1313.2, and 1314.8 nm — a 1.6 nm spacing narrower than the standard CWDM grid. It is therefore better described as a coarse-spaced WDM design, chosen to fit four channels within the available simulation bandwidth while keeping separation wide enough to avoid significant inter-channel crosstalk, rather than the formal ITU-T CWDM standard.

2.2. Photodetector Theory: PIN versus APD

The receiver's photodetector converts the optical signal into an electrical current; its noise characteristics set the receiver sensitivity. Two photodetector technologies are compared in this study.

2.2.1. PIN Photodiode

A PIN photodiode has a lightly doped intrinsic (I) layer between p-type and n-type regions. Absorbed photons generate electron-hole pairs swept out by reverse bias to produce a photocurrent proportional to incident power. With no internal gain, its output is limited by responsivity, and its main noise sources are thermal (Johnson) noise and shot noise. Because of this, PIN photodiodes perform best at relatively high received power, since at low power the fixed thermal noise floor of the receiver circuit dominates and limits sensitivity.

2.2.2. Avalanche Photodiode (APD)

An APD adds a high-field multiplication region where impact ionization creates additional carrier pairs, providing an internal gain M before electronic amplification. This lets the APD detect weaker signals than a PIN photodiode under the same thermal noise conditions. However, the avalanche process adds its own shot noise, quantified by the excess noise factor F(M), which increases with M. As a result, APD performance does not improve indefinitely with gain — at high received power, the added multiplication noise can outweigh the sensitivity benefit, making a PIN photodiode the better choice in that regime.

2.2.3. Expected Trade-off

Based on this theory, the PIN photodiode is expected to outperform the APD at short fiber lengths, where power is high and thermal noise dominates, while the APD is expected to outperform the PIN photodiode at longer lengths, where reduced received power makes its internal gain advantageous. The distance sweep in Section 4 tests this hypothesis and identifies the approximate crossover distance.

2.3. Performance Metrics

- Bit Error Rate (BER): ratio of incorrectly received bits to total bits transmitted; lower is better, with BER $< 1\times10^{-9}$ is commonly accepted as the performance threshold for conventional optical communication systems.
- Q-Factor: a measure of signal quality at the decision circuit, related to BER by $Q \approx 6 \leftrightarrow BER \approx 1\times10^{-9}$; higher Q indicates a cleaner, more reliable signal.
- Eye Diagram: superposition of all received bit periods, used to assess distortion, noise, jitter, and inter-symbol interference; a wide, open eye indicates good signal quality.

SYSTEM ARCHITECTURE AND COMPONENTS

The simulated system consists of a four-channel WDM transmitter implemented using CWDM components, a shared amplified optical fiber link, and a four-channel receiver bank, arranged as a modular optical communication chain in OptiSystem. The complete system block diagram is shown in Figure 1.

3.1. System Block Diagram

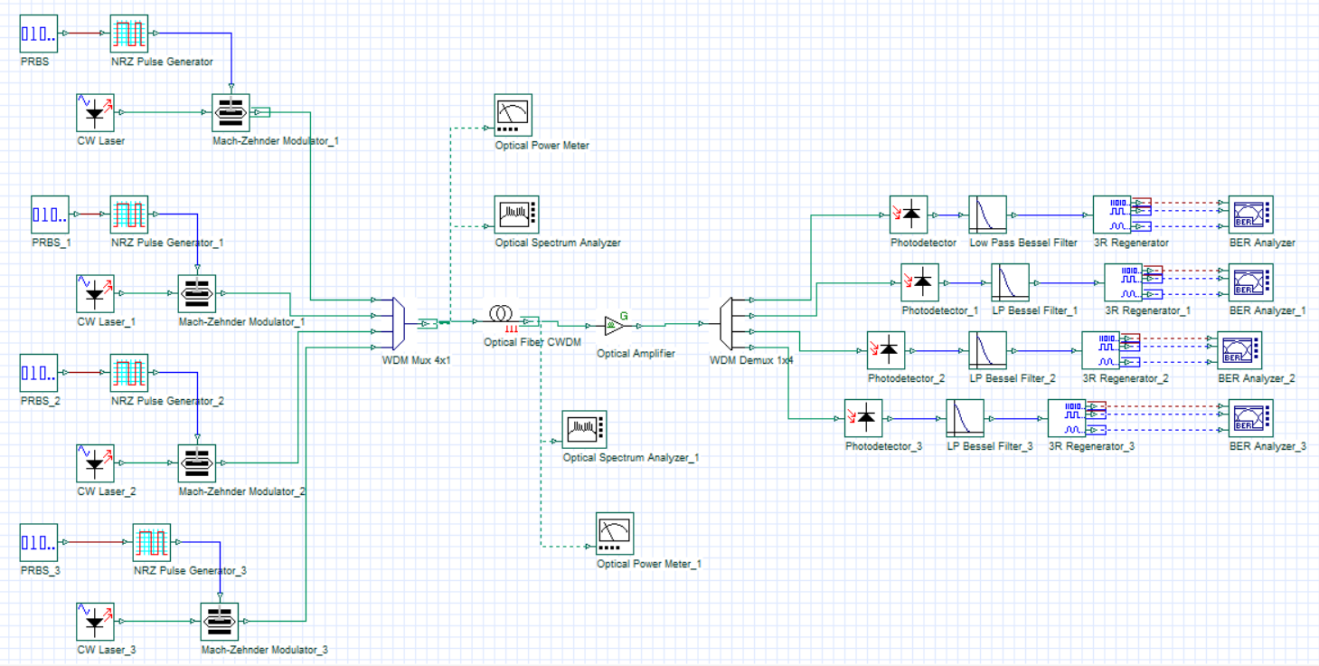


Figure 1: Layout of the simulated 4-channel CWDM system

3.2. Component Description

Table 1 summarizes the function and key configuration parameters of each component used in the system.

Table 1: System component summary

Component	Function	Key Parameter(s) Used
PRBS Generator (×4)	Generates the pseudo-random binary data sequence for each channel	Bit rate = 10 Gbps
CW Laser (×4)	Provides the optical carrier for each WDM channel	1310 / 1311.6 / 1313.2 / 1314.8 nm, Power = 0 dBm

Component	Function	Key Parameter(s) Used
Mach-Zehnder Modulator (×4)	Intensity-modulates the optical carrier with the electrical data signal	Extinction ratio = 30 dB
WDM Mux 4x1	Combines the four modulated channels onto a single fiber	Bandwidth = 1.6 nm, Filter order = 2
Optical Fiber (CWDM type)	Transmission medium between transmitter and receiver	Length = 0–100 km (swept), Attenuation = 0.2 dB/km
Optical Amplifier (EDFA)	Partially compensates for fiber span loss and component insertion loss	Gain = 15 dB, Noise Figure = 4 dB
WDM Demux 1x4	Separates the combined signal back into individual channels	Bandwidth = 1.6 nm, Filter order = 2
Photodetector (PIN / APD)	Converts the optical signal back to an electrical signal	Responsivity = 1 A/W; APD gain (M) = 3 (APD case only)
Low Pass Bessel Filter (×4)	Suppresses out-of-band noise and limits inter-symbol interference before regeneration	Cutoff frequency = 0.75 × Bit rate
3R Regenerator (×4)	Reshapes, retimes, and reamplifies the recovered electrical signal	Reference bit rate = 10 Gbps
BER Analyzer (×4)	Computes Q-factor, BER, eye height, and renders the eye diagram	—

3.3. Global Simulation Parameters

Table 2 lists the global simulation settings used throughout this study, kept identical for both the PIN and APD configurations to ensure a fair comparison.

Table 2: Global simulation parameters

Parameter	Value	Units
Bit rate per channel	10	Gbps
Sequence length	256	Bits
Samples per bit	64	—
Sample rate	640	GHz
Number of channels	4	—

PERFORMANCE OBSERVATION AND ANALYSIS

This section presents the simulation results in six parts: baseline system validation, 50 km power verification, receiver filter evaluation, PIN and APD distance sweep analysis, and detector performance comparison. Channel-to-channel variation across the CWDM band is also examined.

4.1. System Validation

Before performing the full distance sweep, the system was validated at a selected reference fiber length of **20 km**. The validation process included spectral analysis at both the transmitter and receiver, followed by BER and Q-factor measurements. This ensured that the WDM Mux/Demux, optical amplifier, and receiver were correctly configured before conducting the performance comparison between the PIN and APD photodetectors.

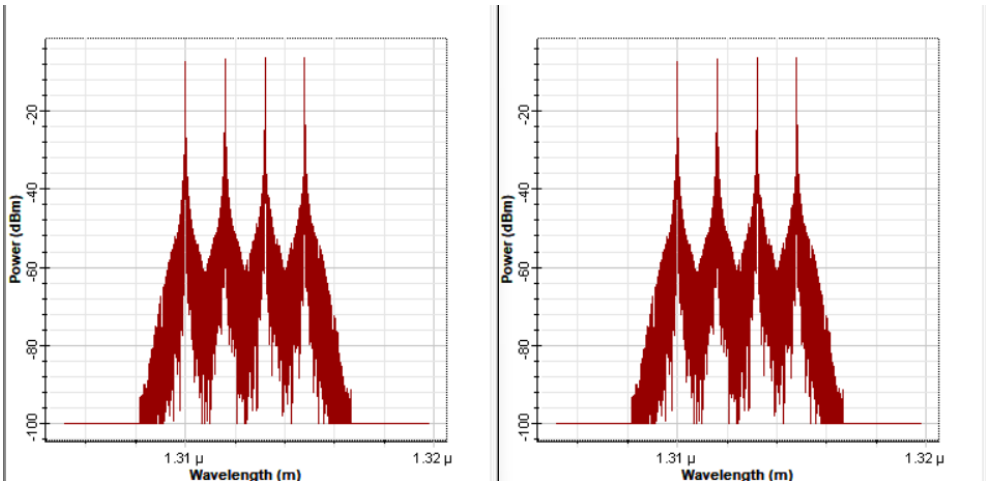


Figure 2: Optical spectrum at the transmitter and receiver (left to right)

The transmitter spectrum confirms the successful multiplexing of the four optical carriers at their assigned wavelengths with clear channel separation and no significant spectral overlap. The received spectrum shows that all four wavelength channels remain distinguishable after transmission, indicating successful propagation through the optical link and adequate amplification to partially compensate for fiber attenuation.

Following spectral validation, the system performance was evaluated using BER and Q-factor measurements for all four channels. The results are summarized in Table 3.

Table 3: Baseline validation results for all four channels at the reference fiber length

S.N.	Channel	λ (nm)	Detector	Max Q Factor	Eye Height	Min. BER
1	1	1310	PIN	11.8178	0.00945607	1.57782e-032
2	1	1310	APD	11.0214	0.0278348	1.50627e-028
3	2	1311.6	PIN	8.24381	0.00825701	8.30302e-017
4	2	1311.6	APD	7.59564	0.0237282	1.52879e-014
5	3	1313.2	PIN	8.27095	0.00822187	6.63286e-017
6	3	1313.2	APD	7.90537	0.0243309	1.33079e-015
7	4	1314.8	PIN	11.0304	0.00940444	1.35545e-028
8	4	1314.8	APD	10.6102	0.0277588	1.33342e-026

As shown in Table 3, all four channels achieved Q-factor values above the commonly accepted telecom benchmark of Q = 6 (corresponding to BER ≈ 1×10^{-9}) for both detector types, confirming that the system was correctly configured prior to the distance sweep. The small variation observed between channels is likely due to EDFA gain ripple across the 1310–1315 nm amplification band together with minor asymmetries in the Mux/Demux filter response between the centrally located and edge channels.

4.2. Optical Power Before and After Fiber Transmission

To validate the 50 km attenuation model, optical power was measured at both transmitter and receiver ends with all other simulation parameters unchanged. The transmitter-side power was approximately 1.981×10^{-3} W (2.97 dBm), as shown in Figure 3, while the receiver-side power after 50 km transmission was approximately 100.121 μW (−9.99 dBm), as shown in Figure 4.

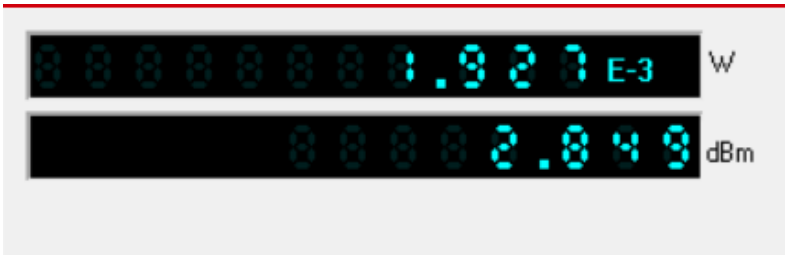


Figure 3: Transmitter-side optical power measurement for the 50 km link

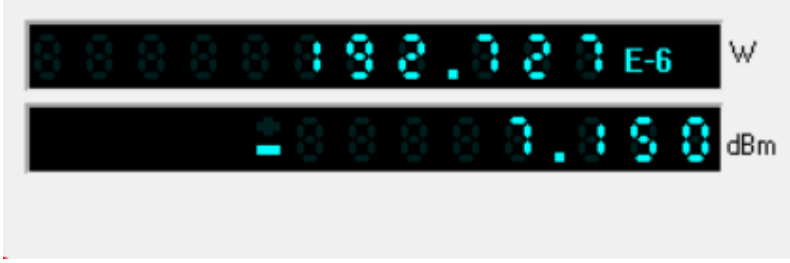


Figure 4: Receiver-side optical power measurement after 50 km fiber transmission

The measured reduction in optical power is approximately 10 dB, which agrees closely with the theoretical fiber attenuation of 50 km × 0.2 dB/km = 10 dB. This agreement validates the attenuation model used in the OptiSystem simulation and confirms that the received-power reduction follows the expected fiber-loss behavior.

4.3. Effect of the Low-Pass Bessel Filter

The effect of receiver-side filtering was investigated for Channel 1 at 50 km by comparing the BER Analyzer results with and without the Low-Pass Bessel Filter. In both cases, the fiber length, transmitter configuration, detector setting, amplifier, demultiplexer, and all other simulation parameters were kept unchanged, so that the observed difference is mainly due to the filter response.

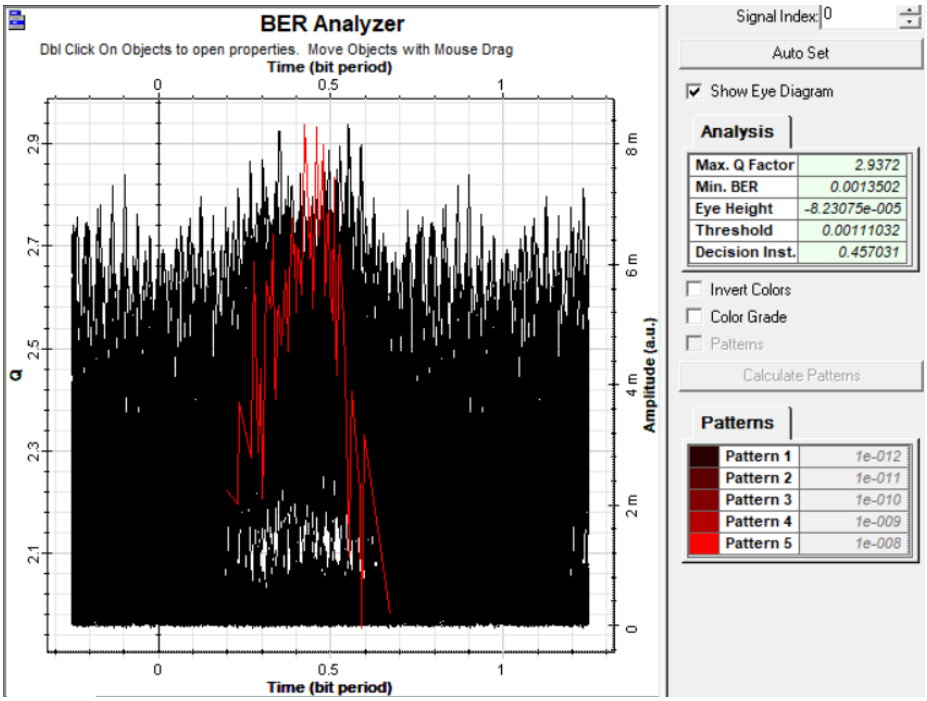


Figure 5: BER Analyzer result for Channel 1 at 50 km without Low-Pass Bessel Filter

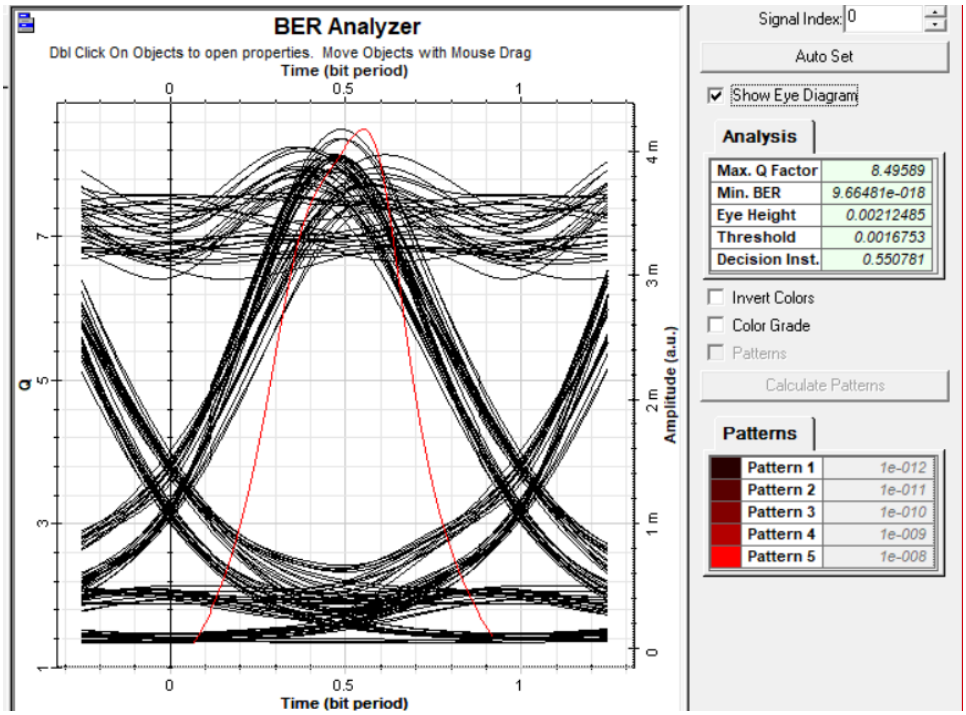


Figure 6: BER Analyzer result for Channel 1 at 50 km with Low-Pass Bessel Filter

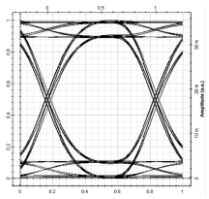
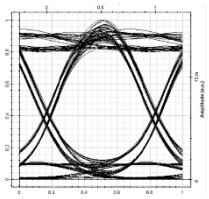
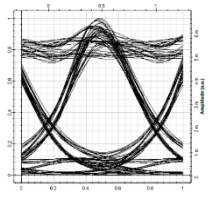
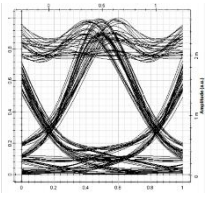
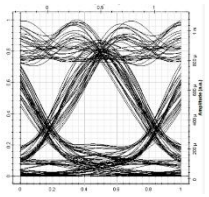
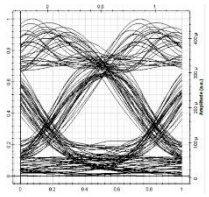
The comparison in Figures 5 and 6 shows that the Low-Pass Bessel Filter significantly improves the received eye opening by suppressing high-frequency noise components and reducing inter-symbol interference before regeneration. Without the filter, the eye diagram is nearly closed and the Q-factor is only about 2.94, with a BER of approximately 1.35×10^{-3}. With the filter included, the eye diagram becomes clearly open, the Q-factor increases to about 8.50, and the BER improves to approximately 9.66×10^{-18}. This confirms that receiver filtering is essential for maintaining signal integrity in the 40 Gbps CWDM system.

4.4. Distance Sweep Analysis

4.4.1. Distance Sweep Analysis of PIN Receiver

To evaluate the impact of transmission distance on system performance, the fiber length was swept from 0 km to 100 km in 10 km increments, using Channel 1 (1310 nm) as the representative channel, with a PIN photodetector at the receiver. It was expected that increasing fiber length would lead to greater attenuation, reducing the received optical power and degrading the Q-factor and eye diagram quality, with degradation accelerating once the receiver's noise floor becomes the dominant limitation.

Table 4: PIN receiver performance versus fiber length (Channel 1, 1310 nm)

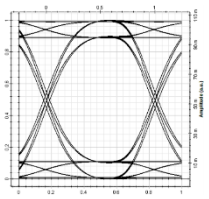
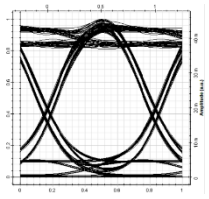
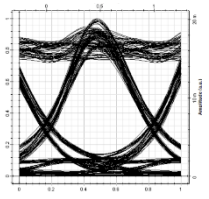
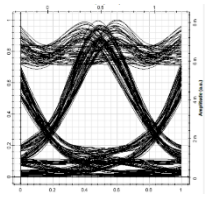
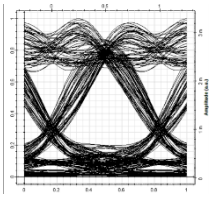
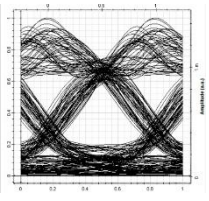
S.N.	Length (km)	Max Q Factor	Eye Height	Min. BER	Eye Diagram
1	0	9.56461	0.0219077	5.63135e-022	
2	20	11.8178	0.00945607	1.57782e-032	
3	40	9.41877	0.00347091	2.28004e-021	
4	60	7.03233	0.00117903	1.00965e-012	
5	80	8.30523	0.000483044	4.9208e-017	
6	100	6.62025	0.000149799	1.78406e-011	

As shown in Table 4, the Q-factor for the PIN receiver decreases overall, peaking at 11.82 at 20 km before falling to 6.62 at 100 km (a drop of approximately 44% from the 20 km peak). This trend is consistent with the expected noise-limited behavior of a PIN photodiode, whose lack of internal gain makes it increasingly sensitive to the reduced received power at longer fiber lengths.

4.4.2. Distance Sweep Analysis of APD Receiver

The identical length sweep (0 to 100 km, Channel 1, 1310 nm) was repeated with the PIN photodetector replaced by an APD, with all other system parameters held constant to ensure a fair comparison.

Table 5: APD receiver performance versus fiber length (Channel 1, 1310 nm)

S.N.	Length (km)	Max Q Factor	Eye Height	Min. BER	Eye Diagram
1	0	8.84261	0.0634479	4.66828e-019	
2	20	11.0214	0.0278348	1.50627e-028	
3	40	8.76071	0.0101788	9.63729e-019	
4	60	6.91187	0.00350437	2.36248e-012	
5	80	8.16142	0.00143719	1.65206e-016	
6	100	6.67921	0.000460038	1.19932e-011	

As shown in Table 5, the Q-factor for the APD receiver also decreases overall, peaking at 11.02 at 20 km before falling to 6.68 at 100 km, a drop of approximately 39% from the 20 km peak, a slightly more gradual decline than the PIN receiver's 44% drop. The APD's internal avalanche gain helps sustain signal quality as the received optical power decreases with increasing fiber length, allowing it to become competitive with and eventually slightly outperform the PIN receiver at the longest transmission distances investigated. This advantage comes at the cost of a slightly lower Q-factor at shorter distances (8.84 versus 9.56 at 0 km), where the excess multiplication noise associated with avalanche gain can outweigh its sensitivity benefit.

4.4.3. PIN versus APD

Figure 7 plots the Q-factor obtained for both the PIN and APD receivers against fiber length.

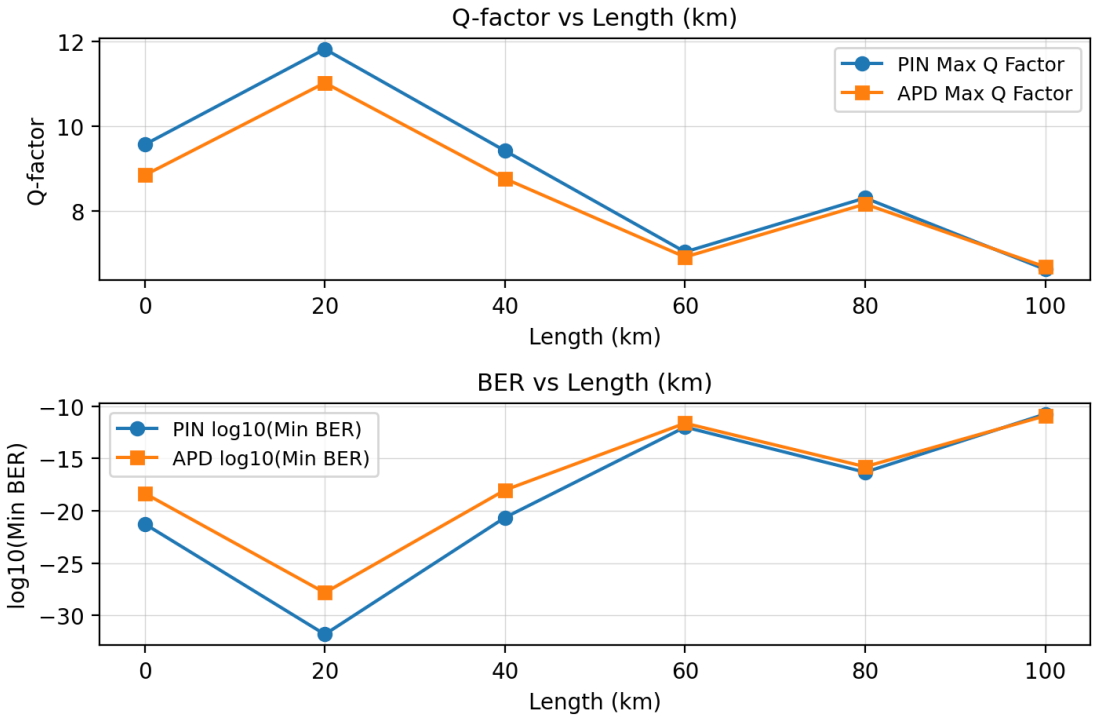


Figure 7: Comparison of PIN and APD receiver performance versus fiber length

The results indicate that the PIN receiver outperforms the APD receiver at shorter fiber lengths, where the received optical power remains relatively high and the PIN photodiode's lower intrinsic noise gives it the advantage in this high received-power regime, without the added avalanche multiplication noise associated with the APD. As fiber length increases and the received power drops due to attenuation, the performance gap between the two receivers narrows, and at the

maximum investigated distance of 100 km the APD receiver exhibits a slight advantage in both Q-factor and BER. This behavior is consistent with the theoretical trade-off discussed in Section 2.2.3: the PIN photodiode's simplicity and lower intrinsic noise favor it at shorter transmission distances, while the APD's internal avalanche gain becomes increasingly beneficial as the received optical power decreases over longer fiber lengths.

Within the 0–100 km range tested for the representative channel (Channel 1), both detector types maintained a BER below the 1×10^{-9} commonly accepted telecom benchmark, with the worst-case value of 1.78×10^{-11} recorded at 100 km for the PIN receiver. However, as discussed in Section 4.5, the center channels (2 and 3) of the CWDM band degrade faster than the edge channels and exceed the 1×10^{-9} benchmark by 100 km, indicating that additional amplification stages or dispersion compensation would be required to extend reliable transmission further for those channels, which is noted as a direction for future work in the Conclusions.

4.5. Channel-to-Channel Variation Across the CWDM Band

To examine whether performance varies systematically across the four CWDM channels, Q-factor and BER were recorded for all four channels at three representative fiber lengths (0, 50, and 100 km) using the PIN receiver.

Table 6: Channel-to-channel Q-factor and BER variation at selected fiber lengths

S.N.	Length (km)	Channel	λ (nm)	Max Q Factor	Min. BER
1	0	1	1310	9.56461	5.63135e-022
2	0	2	1311.6	6.71034	9.70219e-012
3	0	3	1313.2	6.59337	2.14915e-011
4	0	4	1314.8	9.35827	4.05134e-021
5	50	1	1310	8.49589	9.66481e-018
6	50	2	1311.6	6.73486	7.91453e-012
7	50	3	1313.2	6.82334	4.29736e-012
8	50	4	1314.8	8.00805	5.58504e-016
9	100	1	1310	6.62025	1.78406e-011
10	100	2	1311.6	5.04891	2.22071e-007
11	100	3	1313.2	5.15356	1.27458e-007
12	100	4	1314.8	6.14566	3.97003e-010

The results in Table 6 show that the edge channels (1 and 4) consistently outperform the center channels (2 and 3) at all three tested lengths — for example, at 0 km the edge channels average a Q-factor of 9.46 versus 6.65 for the center channels (about 42% higher), and at 100 km the edge channels average 6.38 versus 5.10 for the center channels (about 25% higher). This variation is likely due to a combination of EDFA gain ripple across the amplification band and minor differences in the Mux/Demux filter response for channels located near the center and edges of the multiplexed spectrum. Such variation is commonly observed in practical WDM systems and does not indicate a fault in the simulated design.

CONCLUSION

This study designed, simulated, and validated a four-channel CWDM optical communication system operating at 40 Gbps using OptiSystem, and conducted a detailed comparison of PIN and APD photodetector performance across a fiber length sweep from 0 to 100 km. The baseline system validation confirmed correct operation across all four channels, with Q-factor values of 7.60 to 11.82 achieved at the selected reference fiber length, corresponding to a Bit Error Rate well within the commonly accepted telecom benchmark of 1×10^{-9}.

The distance sweep comparison between PIN and APD receivers confirmed the theoretical expectation that PIN photodiodes perform better at shorter transmission distances due to their lower intrinsic noise, while the APD's internal avalanche gain becomes increasingly beneficial as the received optical power decreases. Based on the simulation results in Section 4.4, the PIN receiver achieved better overall performance over most of the investigated transmission distances, whereas the APD receiver exhibited a slight performance advantage at the maximum investigated distance of 100 km in terms of both Q-factor and BER.

These findings provide practical guidance for photodetector selection in CWDM-based network deployments: PIN receivers are recommended for short-reach metro or access links where cost and simplicity are prioritized, while APD receivers are recommended for extended-reach CWDM links where maintaining link margin over longer distances is the primary concern. Additionally, the channel-to-channel variation observed across the CWDM band highlights the importance of accounting for amplifier gain ripple and filter edge effects when designing multi-channel WDM systems, an aspect that single-channel link studies cannot capture.

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